

# ZIGBEE For Multiple Device Control Using A Single Remote

Today's homes have as many remotes as the number of devices to control. Zigbee wireless technology does away with the need for separate remotes by enabling a standard-based home-area network that doesn't require line-of-sight communication. Here is how ZigBee works and what it has in store for you



■ MANUJ ARORA, VINEET ARORA

**Z**igBee is a home-area network device designed specifically to replace individual remote controls. It satisfies the market's need for a cost-effective, standard-based wireless network that supports low data rates, low power consumption, security and reliability. A smart home and smart office with flexibility and seamless mobility, all without wires, are some of the promises of ZigBee wireless solutions.

The whimsical name of ZigBee comes from the 'Zigzag principle,' i.e., the technique that honey bees use to communicate new-found food sources to other members of the colony. This technique of honey bees is referred to as Waggle dance.

## Technical attributes

ZigBee is based on IEEE 802.15.4 standard for packet-based wireless data transport. This standard addresses communication needs of wireless applications that have low data rates and low power consumption requirements.

To meet the global regulatory requirements, ZigBee standard supports the following unlicensed radio bands: 2.4GHz (worldwide), 868MHz (Europe) and 915MHz (America). Moreover, it has a remarkable range of up to 75 metres.

**Networking overview.** ZigBee network uses three types of devices: ZigBee coordinator, ZigBee router